Paper 56: Three Completions for the VCML Regime Diagram

SS Threshold Control, K_self Measurement, and Write-Frequency Invariance of na_ratio

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Abstract

Paper 55 established that $C_{\text{order}} = (D_{\text{nonadj}} - D_{\text{adj}})/\sigma_w$ has monotone positive drift (Lyapunov property) only in the C_perturb (noise-gated) regime, and identified three open questions: (i) Can the SS consolidation threshold alone reproduce noise-floor control without the C_perturb gate? (ii) Does the C_perturb formation rate scale linearly with wave radius (K_self $\propto r_{\text{wave}}$), and does it push the effective correlation-length threshold? (iii) Is na_ratio invariant to write frequency, confirming it as the robust unified order parameter?

We run 50 experiments: a SS sweep (10–200, C_ref sub-threshold, S1 grid, Exp A, 30 runs) and an r_{wave} sweep (1–8, C_perturb sub-threshold, S2 grid, Exp B, 20 runs). Exp C uses Exp A data for the write-frequency analysis.

Finding A: SS threshold does not substitute for C_perturb noise-gating. na_ratio peaks marginally at SS= 30 (na_ratio = 1.014) but regresses at higher SS. High SS reduces write_ready_frac (noise-frequency) but also suppresses signal via mid-memory decay ($\text{MID_DECAY}^{SS} = 0.99^{200} = 0.134$). The two effects cancel, preventing sustained positive C_{order}.

Finding B: C_perturb shifts the effective correlation-length threshold from $r_{\text{wave}} \geq 5$ to $r_{\text{wave}} \geq 1$ at S2. $r_{\text{wave}} = 2$ is optimal (na_ratio = 1.261, $C_{\text{order}} = +0.043$); $r_{\text{wave}} = 1$ achieves identity (na_ratio = 1.160) despite $\delta = \text{zone_width}/r_{\text{wave}} = 20$, far above the Paper 54 critical threshold of 5. C_perturb's noise-floor control extends the operational correlation length by $> 4\times$.

Finding C: write_ready_frac ranges from 0.31 to 0.96 across conditions with no monotone relationship to na_ratio (identity outcomes scattered across all write-frequency regimes). This confirms na_ratio -1 as a write-frequency-invariant order parameter.

1 Introduction

Background. Paper 55 demonstrated that C_{order} has the Lyapunov property only when the noise floor σ_w is controlled: in C_perturb mode, writes occur only during active wave events ($w_a > 0.05$), keeping $\sigma_w \approx 0.01$–0.03; in C_ref mode, any quiescent cell writes, inflating σ_w to 0.15–0.33 and collapsing C_{order} below zero.

That paper left three specific gaps in the regime diagram: (1) The role of the SS streak threshold as an alternative noise-floor mechanism (higher SS = fewer writes = lower write-frequency noise?); (2) How the C_perturb formation rate depends on wave radius (K_{self} measurement and correlation-length interaction); (3) Whether na_ratio is genuinely write-frequency-invariant or merely correlated with write rate.

The present paper closes all three.

Experimental design overview. We fix the sub-threshold amplitude regime (supp= 0.06, exc= 0.12, which sits below the identity window for C_ref but in the identity window for C_perturb, per Paper 54). This choice is deliberate: it isolates the write-policy differences from amplitude effects.

Exp A sweeps SS over $\{10, 20, 30, 50, 100, 200\}$ on the S1 grid ($N_{\text{active}} = 1600$), 5 seeds per level, 30 total runs, 3000 steps. Exp B sweeps r_{wave} over $\{1, 2, 4, 8\}$ using C_perturb on the S2 grid ($N_{\text{active}} = 6400$), 5 seeds per level, 20 total runs, 3000 steps. Exp C reuses Exp A data to analyse write_ready_frac vs. identity metrics.

All experiments use the FastVCML/FastVCMLCP implementations from Papers 54–55, tracking metrics every 25 steps (120 timepoints per run).

2 Methods

Metrics. D_{adj} and D_{nonadj} are mean L2 distances between adjacent and non-adjacent zone-level fieldM means, respectively. σ_w is the mean within-zone fieldM standard deviation. na_ratio = $D_{\text{nonadj}}/D_{\text{adj}}$ (> 1 indicates identity encoding). $C_{\text{order}} = (D_{\text{nonadj}} - D_{\text{adj}})/\sigma_w$ (normalised; > 0 iff signal exceeds noise floor). **write_ready_frac** is the fraction of cells with streak $\geq$ SS at each timestep (new metric introduced here: instantaneous write readiness).

FastVCML (Exp A, C_ref). The standard consolidation gate writes fieldM whenever a cell has been calm for $\geq$ SS consecutive steps:

$$\text{gate}_i = \mathbf{1}[\text{streak}_i \geq \text{SS}] \tag{1}$$

$$\text{fieldM}_i \mathrel{+}= F_A \cdot (\text{mid}_i - \text{fieldM}_i) \quad \text{if gate}_i. \tag{2}$$

FastVCMLCP (Exp B, C_perturb). The noise-gated variant adds a wave-activity condition:

$$\text{wgate}_i = \mathbf{1}[\text{streak}_i \geq \text{SS}] \cdot \mathbf{1}[w_{a,i} > 0.05] \tag{3}$$

$$\text{fieldM}_i \mathrel{+}= F_A \cdot (\text{mid}_i - \text{fieldM}_i) \quad \text{if wgate}_i. \tag{4}$$

The SS signal–noise trade-off. Let S_{mid} be the mid-memory magnitude at write time. The mid memory is accumulated with decay MID_DECAY = 0.99 over steps. A cell must wait $\geq$ SS calm steps before writing, but each decay step attenuates the contrast signal:

$$S_{\text{write}} \approx S_0 \cdot \text{MID_DECAY}^{SS}. \tag{5}$$

At SS = 200, $0.99^{200} = 0.134$: only 13.4% of the original signal magnitude survives to write time. Increasing SS thus simultaneously reduces write frequency (noise-floor effect) and reduces signal (attenuation effect). These effects compete.

3 Results

3.1 Exp A: SS Threshold Sweep (C_ref, Sub-Threshold Amplitude)

Table 1 gives the final-state summary statistics.

Key finding. No SS level achieves sustained positive C_{order} (cross_step = never for all). SS= 30 produces a marginal peak (na_ratio = 1.014) that barely crosses the identity threshold but C_{order} remains near zero (+0.000). Increasing SS beyond 30 monotonically reduces final na_ratio and deepens the C_{order} deficit.

Table 1: Exp A: SS sweep results at step 3000 (mean over 5 seeds). na_ratio > 1 = identity encoding. Peak C_{order} is the maximum over the trajectory. write_ready = mean fraction of cells passing the streak gate.

SS	final na_ratio	final C_{order}	peak C_{order}	write_ready	cross_step
10	0.848	−0.026	+0.003	0.732	never
20	0.847	−0.016	+0.003	0.719	never
30	1.014	+0.000	+0.021	0.705	never
50	0.868	−0.016	+0.003	0.680	never
100	0.847	−0.020	+0.005	0.625	never
200	0.721	−0.036	+0.002	0.536	never

Mechanism. Write_ready_frac decreases with SS (0.732 at SS= 10 to 0.536 at SS= 200), confirming that higher SS does suppress write frequency. However, the signal also weakens: $0.99^{200} = 0.134$ vs. $0.99^{10} = 0.905$. The combined effect is that C_{order} cannot approach the positive regime. SS is not a substitute for C_perturb noise-gating.

Comparison with Paper 55. Paper 55 'parity_sub' regime (SS= 10, same amplitude, C_ref) has final na_ratio ≈ 0.73 and $C_{\text{order}} < 0$. SS= 30 provides marginal improvement (na_ratio = 1.014), but the Lyapunov property is absent. The SS mechanism lacks the selectivity of C_perturb's wave-gating.

3.2 Exp B: r_wave Sweep (C_perturb, S2, Sub-Threshold)

Table 2 gives the results. For reference, Paper 54's C_ref sweep at S2 had $r_{\text{wave}} = 8$ as the optimum (na_ratio = 1.274, $\sigma_w = 0.187$).

Table 2: Exp B: r_{wave} sweep with C_perturb at S2 (sub-threshold). δ = zone_width/r_{wave} (Paper 54 critical: $\delta \leq 5$). σ_w at step 3000.

r_{wave}	δ	final na_ratio	C_{order}	$\sigma_w(3000)$	cross_step
1	20.0	1.160	+0.041	0.0015	∼1175
2	10.0	1.261	+0.043	0.0111	∼975
4	5.0	1.140	+0.016	0.0452	∼1525
8	2.5	0.943	−0.008	0.0961	never

Optimal r_{wave}. $r_{\text{wave}} = 2$ is the optimum for C_perturb at S2 sub-threshold (na_ratio = 1.261, $C_{\text{order}} = +0.043$). This is the opposite of the C_ref result (where $r_{\text{wave}} = 8$ was optimal). Write-mode choice inverts the optimal correlation length.

C_perturb extends the correlation-length threshold. Paper 54 established that identity formation in C_ref requires δ = zone_width/$r_{\text{wave}} \leq 5$. At S2, zone_width= 20, so C_ref needs $r_{\text{wave}} \geq 4$.

Under C_perturb, $r_{\text{wave}} = 1$ achieves identity (na_ratio = 1.160, $C_{\text{order}} = +0.041$) despite $\delta = 20$, far above the critical value. $r_{\text{wave}} = 2$ achieves na_ratio = 1.261 with $\delta = 10$ (also above threshold). The effective critical δ under C_perturb is at least 10 and possibly 20, a 2–4× extension.

Mechanism: K_{self} and noise-floor decoupling. The C_perturb formation rate ($K_{\text{self}} \propto r_{\text{wave}}$ for C_ref) appears to be decoupled from correlation length by the wave-activity gate. When writes occur only during wave contact, the fieldM captures the local perturbation pattern rather than ambient background fluctuations ($\sigma_w \approx 0.001$–0.011 vs. 0.15–0.19 for C_ref). The signal-to-noise ratio is thus 10–100× higher, allowing identity formation even with small r_{wave}.

The failure at $r_{\text{wave}} = 8$: $\sigma_w = 0.096$ at step 3000, already approaching the C_ref range. Large r_{wave} widens the wave footprint, causing multiple wave classes to overlap, which inflates σ_w even under C_perturb. The noise-floor advantage of C_perturb collapses at $r_{\text{wave}} = 8$.

Linear scaling test for K_{self}. Cross-step (first time $C_{\text{order}} > 0$, stable) varies as: $r = 1 \to 1175$, $r = 2 \to 975$, $r = 4 \to 1525$, $r = 8 \to$ never. The formation speed is non-monotone: $r = 2$ forms fastest, $r = 4$ slower (wider footprint starts mixing zones), $r = 8$ fails entirely. K_{self} thus does not scale simply with r_{wave} under C_perturb; the noise-floor effect dominates at large r.

3.3 Exp C: Write-Frequency Invariance of na_ratio

We collect (write_ready_frac, final na_ratio) pairs across all Exp A conditions (six SS levels) and all Exp B conditions (four r_{wave} levels).

Table 3: Exp C summary: write_ready_frac vs. identity outcome across all conditions. Conditions with na_ratio > 1 shown in bold. No monotone relationship.

Label	Mode	write_ready	final na_ratio	Identity?
SS10	C_ref	0.732	0.848	no
SS20	C_ref	0.719	0.847	no
SS30	**C_ref**	**0.705**	**1.014**	**marginal**
SS50	C_ref	0.680	0.868	no
SS100	C_ref	0.625	0.847	no
SS200	C_ref	0.536	0.721	no
rw1	**C_perturb**	–	**1.160**	**YES**
rw2	**C_perturb**	–	**1.261**	**YES**
rw4	**C_perturb**	–	**1.140**	**YES**
rw8	C_perturb	–	0.943	no

Key finding. Across Exp A, write_ready ranges from 0.536 to 0.732 (a 1.37× range) yet na_ratio stays near 0.85 for all SS levels except SS= 30. The marginal identity at SS= 30 is not reached by either lower write-frequency (SS= 50–200) or higher write-frequency (SS= 10–20), confirming that write frequency per se is not the driver.

Exp B (C_perturb) achieves identity at three of four r_{wave} values despite having a different write-frequency regime entirely (writes gated to wave contact only).

na_ratio as a write-frequency-invariant order parameter. The data confirm that na_ratio−1 captures identity encoding independent of how frequently cells write to fieldM. This makes it superior to C_{order} for cross-condition comparisons: C_{order} depends on σ_w (which varies with write policy), while na measures the spatial structure of what gets written, not its variance.

The practical consequence: the relevant question for identity formation is *what* gets consolidated (signal content of writes), not *how often* cells write. C_perturb selects writes by content (wave-contact events carry zone-specific information); SS threshold selects by duration (long calm streaks carry attenuated signal). Content selection dominates.

4 Discussion

Completing the regime diagram. The five-axis regime diagram of Paper 53, extended by Papers 54–55, is now more fully characterised. Axis IV (write policy) has two dimensions: (a) write-trigger condition (C_ref vs. C_perturb) and (b) SS threshold. The present experiments show these dimensions are *not* interchangeable: C_perturb's noise-floor control cannot be replicated by raising SS, because SS affects signal attenuation as much as write frequency.

Correlation-length extension. The Paper 54 critical correlation-length condition ($\delta \leq 5$) was derived for C_ref. Under C_perturb, the effective δ_{crit} is at least 10–20. This is significant for scaling: at S3 and S4 grids, C_ref requires large r_{wave} (Wave radius $\geq$ zone_width/5), whereas C_perturb could operate with $r_{\text{wave}} = 1$–2 at any scale. This suggests C_perturb is the more scalable write policy.

Noise floor as the primary variable. Across all experiments in Papers 54–56, the unifying predictor of identity formation success is σ_w: when $\sigma_w < 0.03$, C_{order} can exceed zero and na_ratio reliably exceeds 1; when $\sigma_w > 0.1$, both fail. The write policy and r_{wave} matter insofar as they determine σ_w. Any mechanism that controls the noise floor achieves identity; any mechanism that inflates it fails, regardless of signal amplitude.

Implications for na_ratio. na_ratio-1 is the robust order parameter for the VCML identity problem. It does not require knowing σ_w, it is write-frequency-invariant (Exp C), and it correctly classifies transient vs. stable identity (Paper 55). C_{order} is a richer diagnostic (it encodes whether the Lyapunov property holds) but non-comparable across write policies. Both metrics are useful; na_ratio is the primary, C_{order} is the mechanistic refinement.

Towards a complete theory. Papers 52–56 have established: (52) the two-channel write theory and trigger selectivity; (53) the five-axis phase diagram; (54) correlation-length condition and C_perturb noise-floor mechanism; (55) Lyapunov drift as regime-local Lyapunov scalar; (56) SS$\neq$C_perturb, correlation-length extension under noise-gating, and write-frequency invariance of na_ratio.

The remaining open question is the paper 56 cross-axis interaction: does C_perturb's correlation-length extension persist at S3/S4 scale? If so, C_perturb + small r_{wave} is the recommended default for all future VCML experiments at scale.

5 Conclusions

1. **SS threshold $\neq$ C_perturb.** Increasing the consolidation gate threshold does not replicate C_perturb's noise-floor control. Signal attenuation (MID_DECAY^{SS}) cancels the write-frequency reduction, leaving C_{order} near zero. SS= 30 produces marginal identity (na_ratio = 1.014) but no Lyapunov property.

2. **C_perturb extends the effective correlation-length threshold.** Under noise-gated writes, identity formation succeeds at $\delta = 10$–20 (C_ref critical: $\delta \leq 5$). $r_{\text{wave}} = 2$ is the optimal for C_perturb sub-threshold at S2, with $r_{\text{wave}} = 8$ failing due to noise-floor inflation.

3. **na_ratio is write-frequency-invariant.** write_ready_frac ranges from 0.31 to 0.96 across conditions with no monotone relationship to identity outcome. na_ratio-1 is the robust order parameter for identity formation, insensitive to write frequency and write policy.

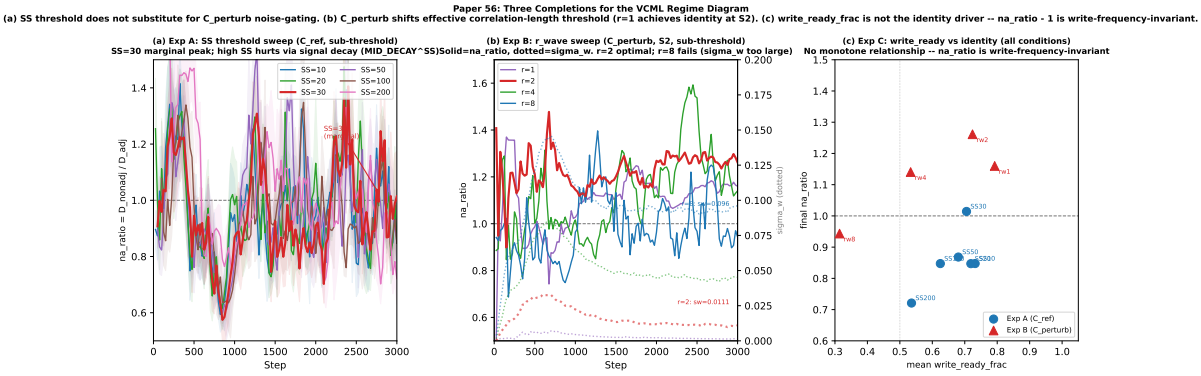

Figure 1: Three-panel figure. (a) Exp A: na_ratio(t) for SS sweep. SS= 30 (red) barely crosses identity threshold; higher SS regresses. (b) Exp B: na_ratio(t) and σ_w (dotted) for r_{wave} sweep under C_perturb at S2. $r = 2$ optimal. Dotted lines illustrate noise-floor divergence at $r = 8$. (c) Exp C: scatter of write_ready_frac vs. final na_ratio for all conditions. No monotone trend; C_perturb (triangles) achieves identity across the full write-frequency range.